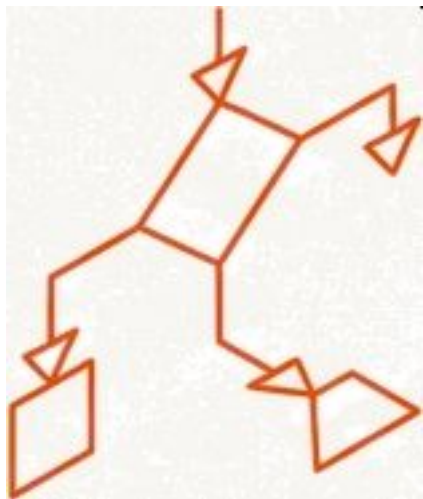




The Scope of Artificial Intelligence

AI is designed to solve distinct classes of problems that require decision-making, pattern recognition, prediction, and learning from experience.



1. Search Problems

Finding a path to a goal (e.g., navigation).



2. Classification Problems

Categorizing data (e.g., image recognition).



3. Prediction Problems

Forecasting future states (e.g., weather or stock trends).



4. Optimization Problems

Finding the best solution among many (e.g., logistics).



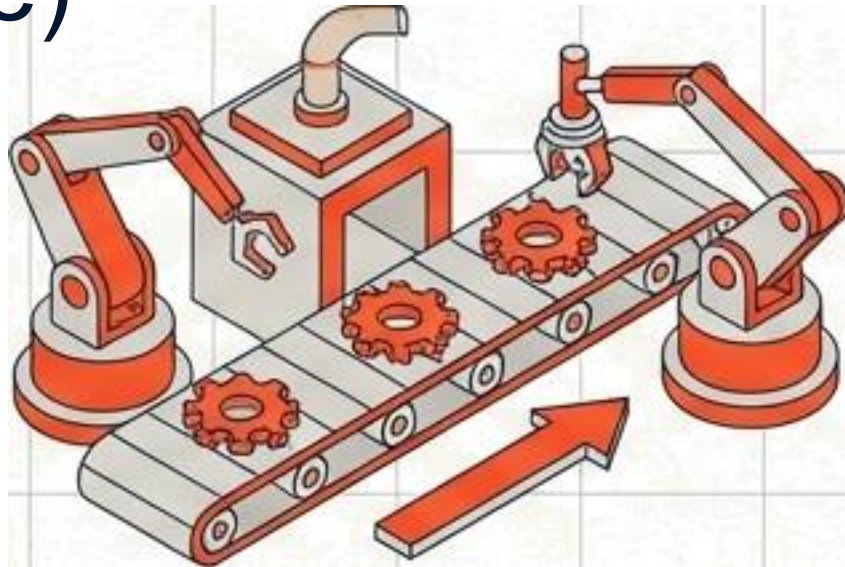
5. Decision-making

Acting with incomplete information under uncertainty.

The Builder's Toolkit: Frameworks & Ecosystems

Modern AI development does not require coding neural networks from zero. Developers utilize sophisticated AI toolkits—software frameworks that handle complex mathematics, optimization, and hardware usage—allowing the focus to remain on the problem and the data.

TensorFlow (Google)



Production

- Focus: Industry-focused and production-ready.
- Best For: Large-scale applications and deployment.

PyTorch (Meta)



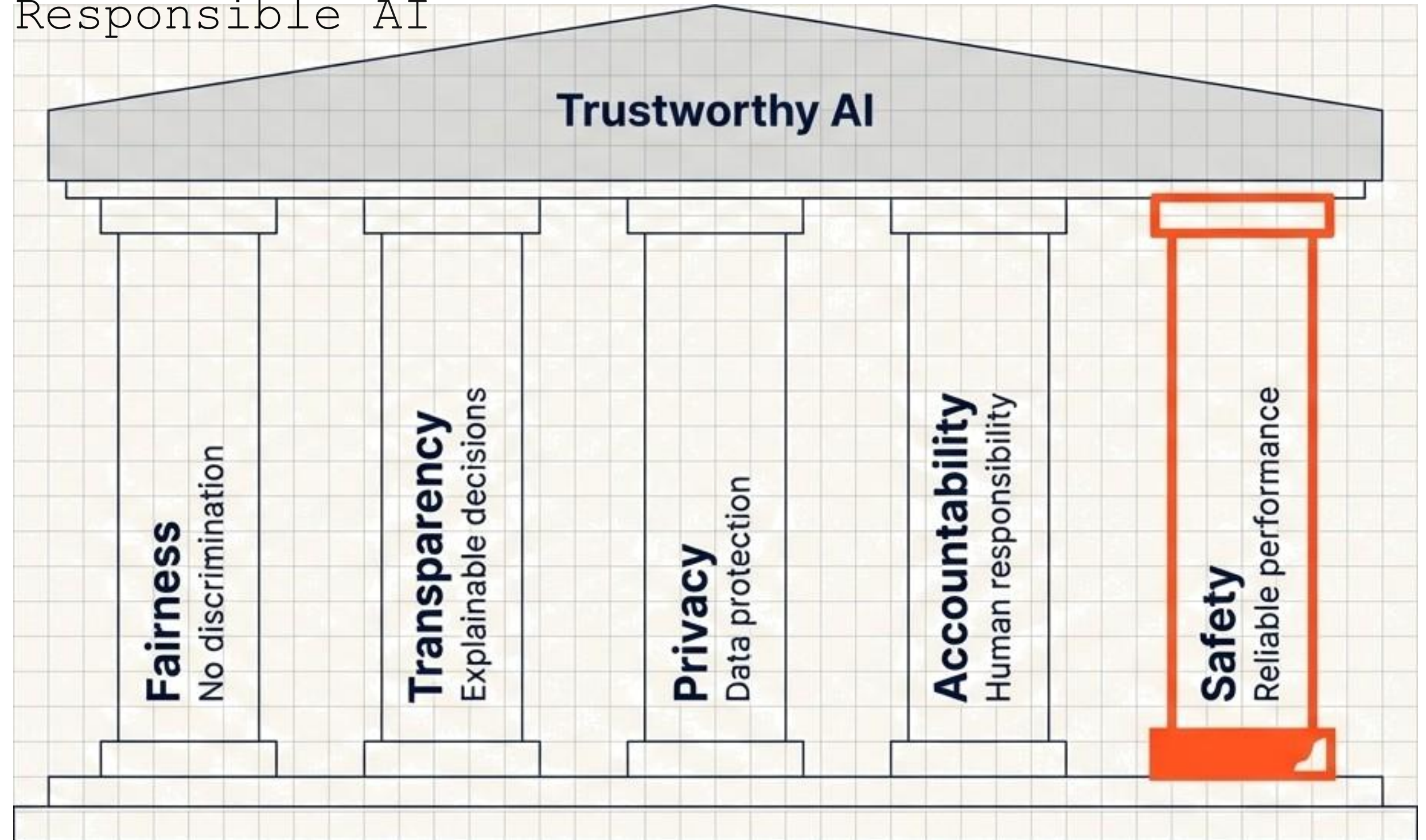
Research &
Experimentation

- Focus: Research-friendly.
- Best For: Experimentation, learning, and easy debugging.

The Ethical Imperative: Responsible AI

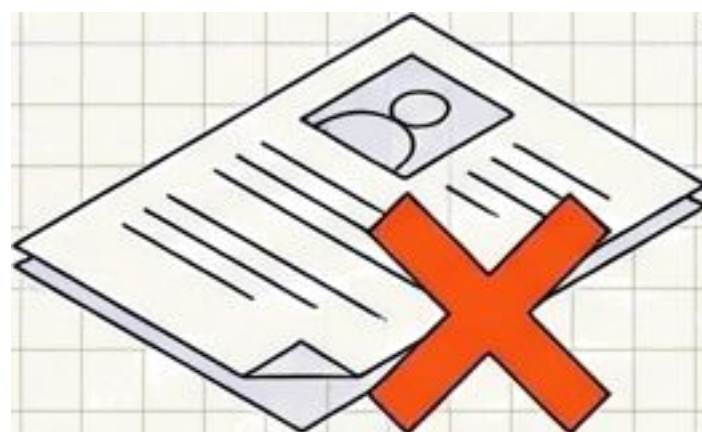
AI systems directly impact people, society, privacy, and fairness. Poorly designed AI can cause tangible harm. Therefore, "Responsible AI" is not optional; it is a core requirement of the architecture.

The 5 Pillars of Responsible AI



Case Study: When AI Fails

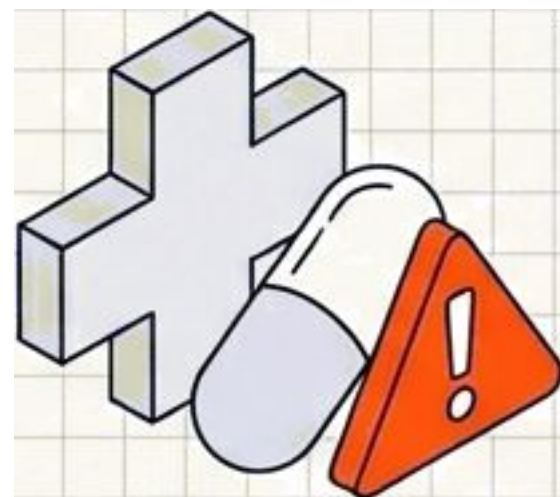
The tangible cost of ignoring ethical constraints.



Biased Hiring Systems

When training data contains historical prejudices, the AI automates discrimination, rejecting qualified candidates based on gender or ethnicity.

VIOLATION: **FAIRNESS**



Medical Prediction Errors

Inaccurate diagnostic models can lead to incorrect treatments, directly harming patient health.

VIOLATION: **SAFETY**

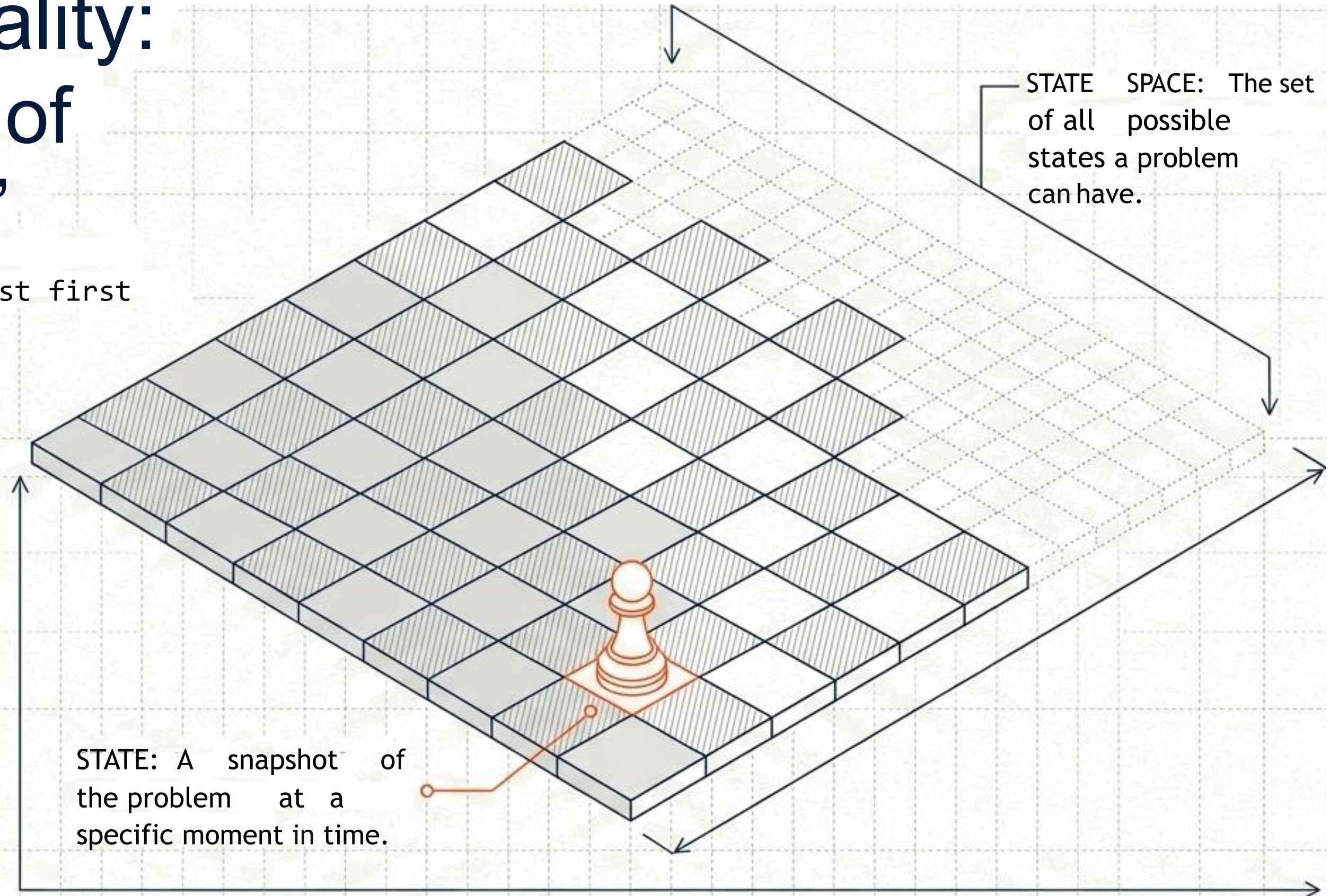
Takeaway: Without the guardrails of Responsible AI, errors can be costly or dangerous.

Modeling Reality: The Concept of "State Space"

To solve a problem, an AI must first map the real world into this structured format.

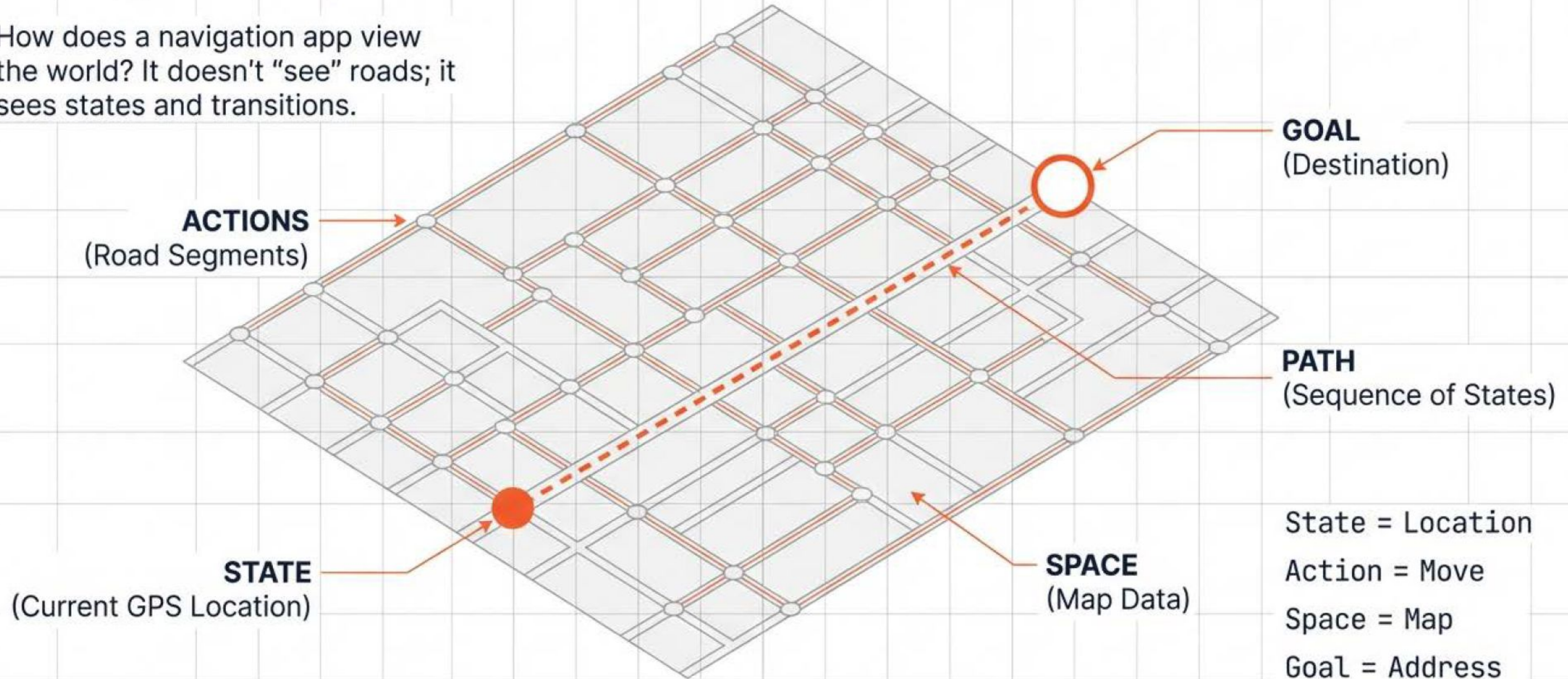
The State Space Model

1. **State:** The exact configuration.
2. **Space:** The universe of all configurations.
3. **Transition:** Moving from one state to another.



Case Study in Modeling: Google Maps

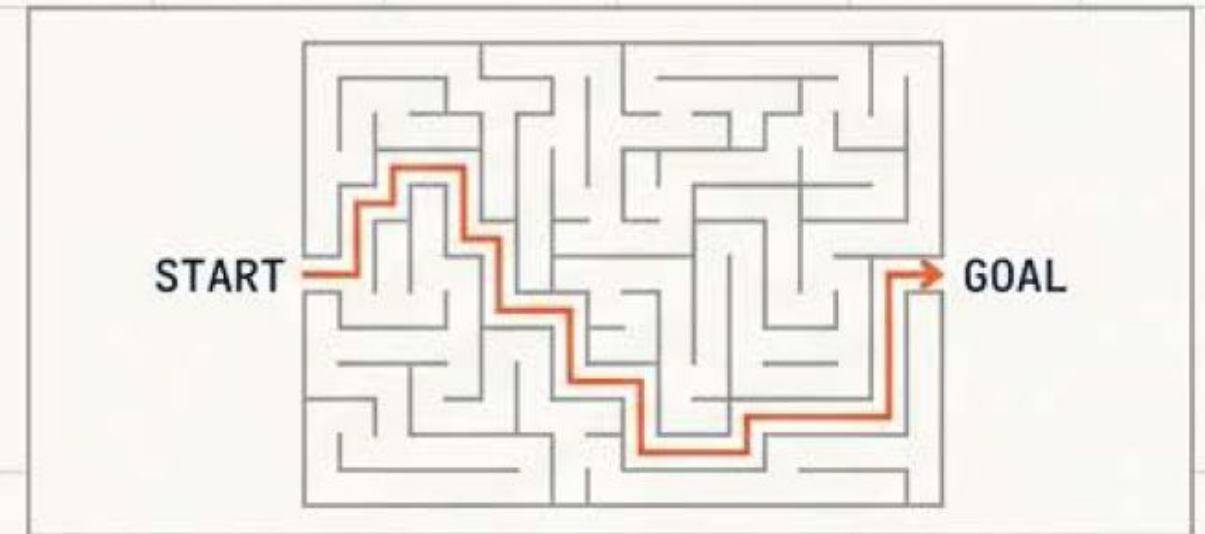
How does a navigation app view the world? It doesn't "see" roads; it sees states and transitions.



The Mechanism: Defining Search Problems

SEARCH PROBLEM: Finding a path from an Initial State to a Goal State through a sequence of valid Actions.

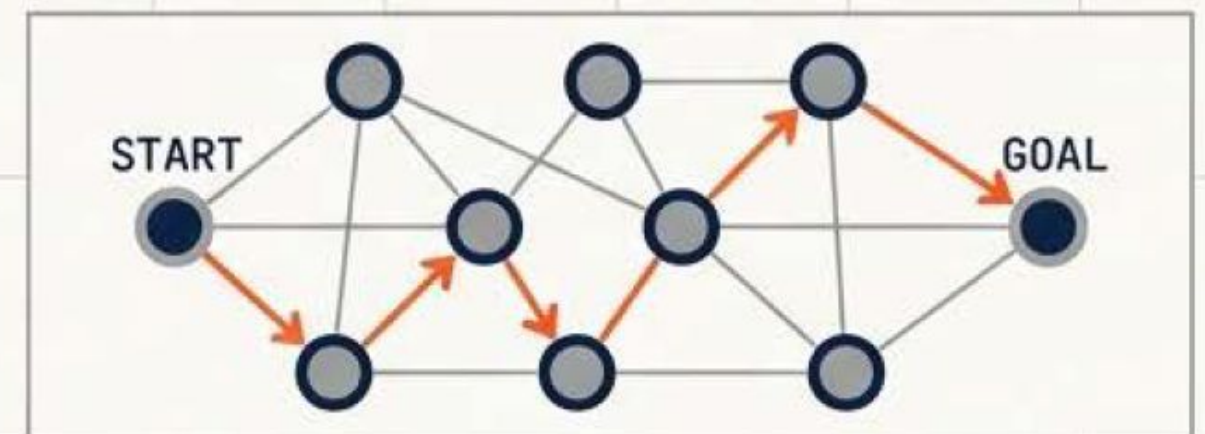
Searching is not just 'looking'—it is the computational process of simulating different options until the goal is reached.



Maze Solving



Puzzle Games



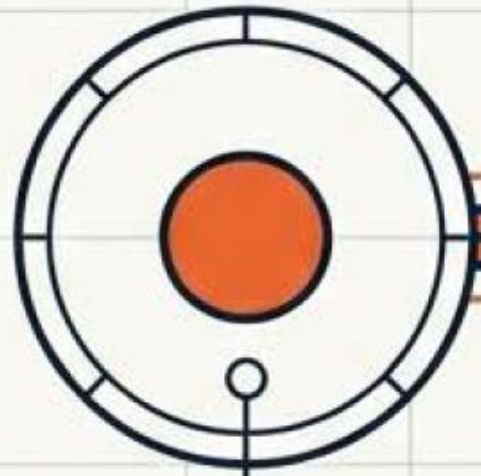
Route Planning

Anatomy of a Search Problem

"All search problems, from simple puzzles to complex logistics, follow this structure."

INITIAL STATE

e.g., Your Location



Where the agent begins.

ACTIONS

e.g., Drive North



Available moves/transitions.

COST

e.g., Time/Distance



Numerical value to optimize efficiency.

GOAL STATE

e.g., Destination

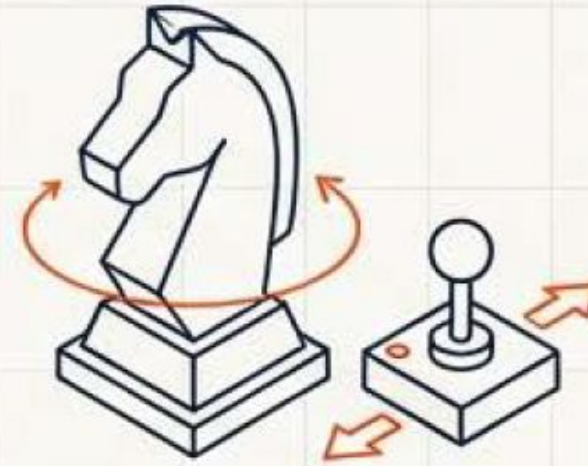


The condition to achieve.

Why It Matters: Real-World Applications

The concept of State Space and Search is the mathematical foundation for intelligent systems across industries. Before building AI, we must first define the problem correctly.

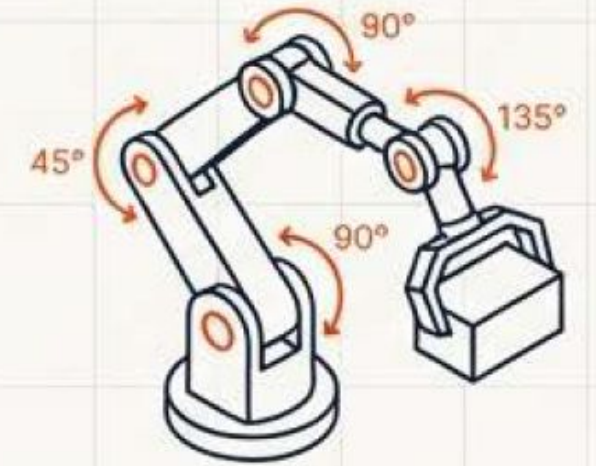
Games



Games

NPCs calculating moves against a player.

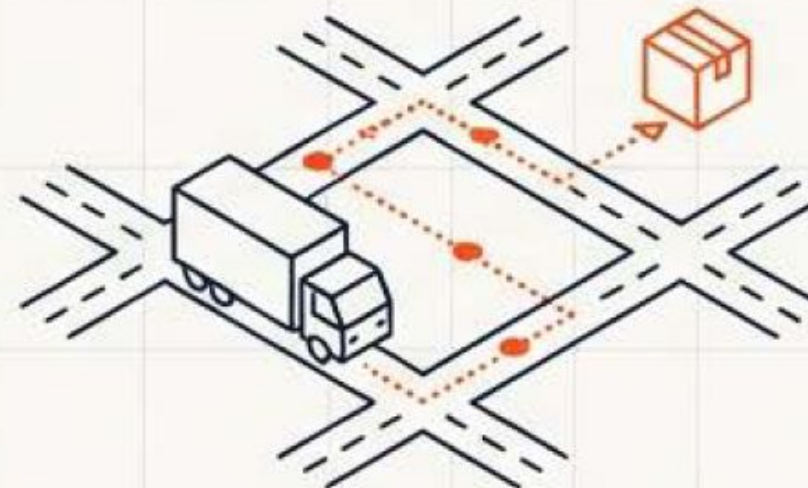
Robotics



Robotics

Determining motion required to grasp an object.

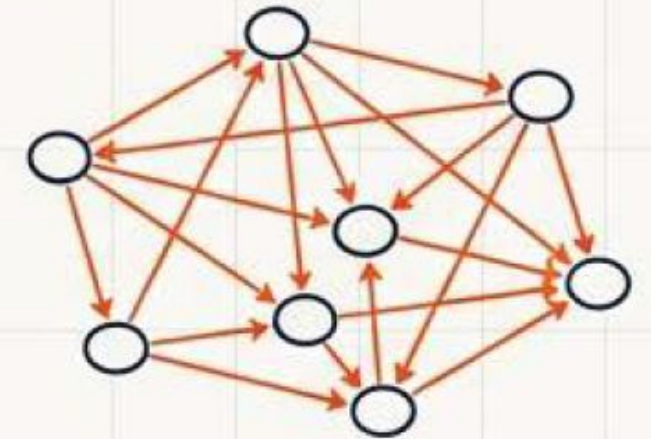
Planning Systems



Planning Systems

Software optimizing delivery routes.

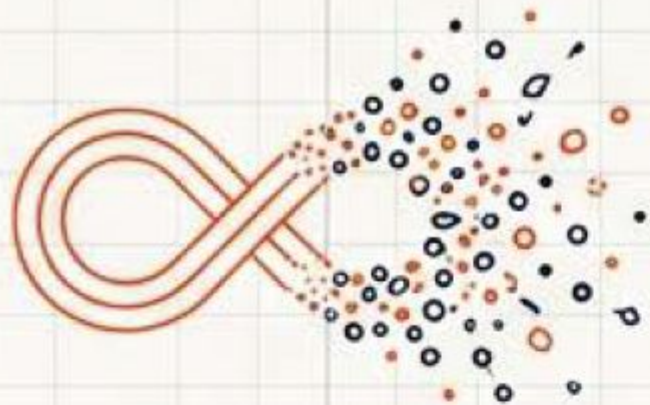
Optimization



Optimization

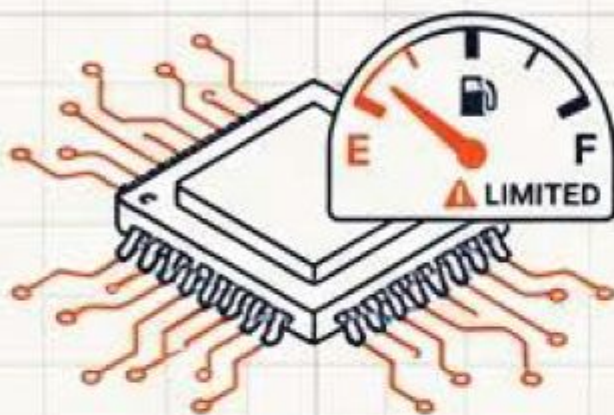
Resource allocation in complex networks.

The Constraints: Challenges in Problem Solving



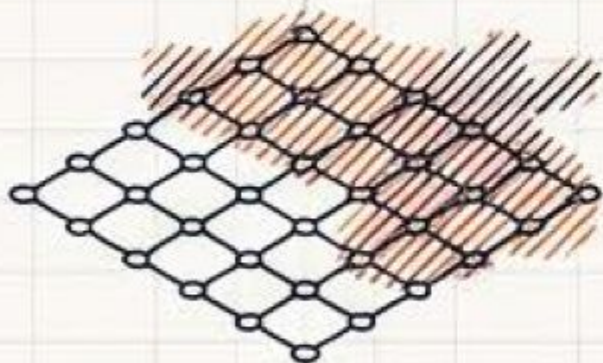
Huge State Spaces

Problems like Go or protein folding have more states than atoms in the universe. Exhaustive search is impossible.



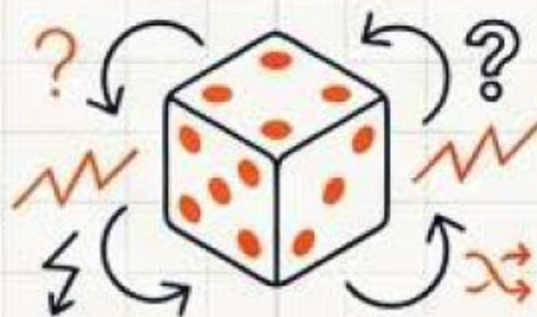
Limited Computation

Hardware resources constrain how deep or fast we can search.



Incomplete Information

The AI may not know the full state (e.g., fog of war, hidden traffic).



Real-World Uncertainty

Unpredictable elements that disrupt the planned path.